# Awesome UI with .NET MAUI





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# David Ortinau

## Principal Product Manager

St. Louis, Missouri USA

Help customers succeed delivering client app experiences across various screens using .NET SDKs for mobile and desktop cross-platform – aka .NET MAUI (and Xamarin).

I like to trail run and have completed multiple distances up to 50 miles.

Grandfather. Doesn't age.



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PROTECTION
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AUSTRIA
AFGHANISTAN
ALBANIA

# Today's List of Awesome UI

Accessibility

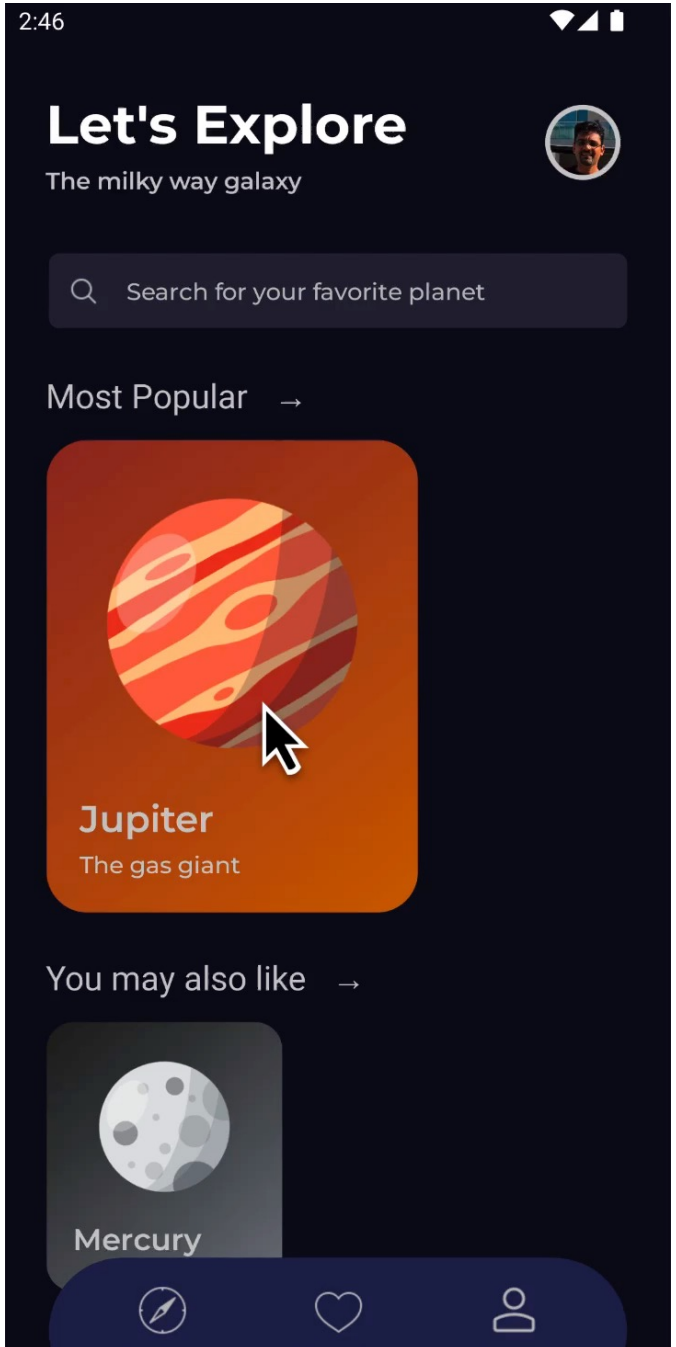
Styles and Themes

Navigation – flyout menu, tabs

Control customizations

State Container

Animations



# Accessibility

# Semantic Properties

## SemanticProperties.Description

Description that will be read aloud by the screen reader

## SemanticProperties.Hint

Provides additional context such as the purpose of a control

## SemanticProperties.HeadingLevel

Enables an element to be marked as a heading to organize the UI and make it easier to navigate

```
<Image Source="logo.png"
       SemanticProperties.Description=".NET logo" />

Image image = new Image { Source = "logo.png" };
SemanticProperties.SetDescription(image, ".NET logo");
```

# Semantic Extensions

## SetSemanticFocus()

Forces screen reader focus to a specified element

## Announce()

Instructs a screen reader to announce text to the user

```
Image image = new Image { Source = ”logo.png” };
Image.SetSemanticFocus();

SemanticScreenReader.Default.Announce(“I am being announced”);
```

# Community Toolkit

## SemanticEffects

Set description, hint, and heading level plus include/exclude.

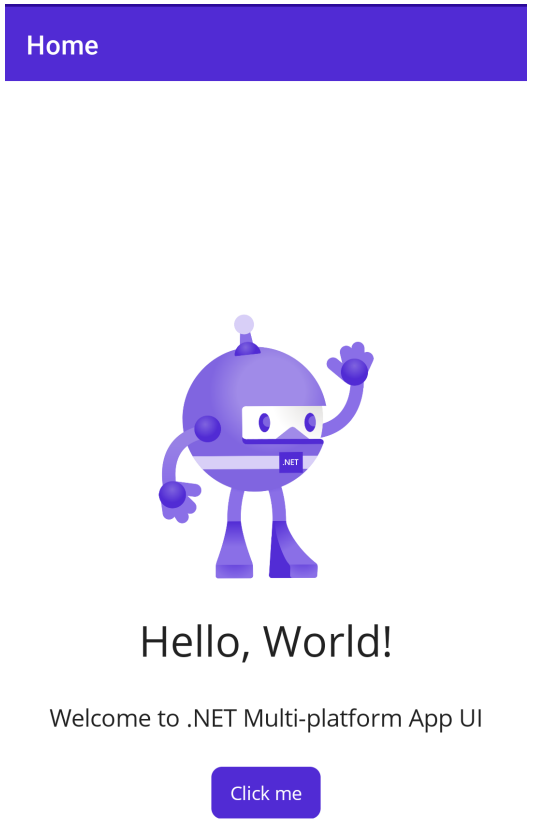
## SemanticOrderView

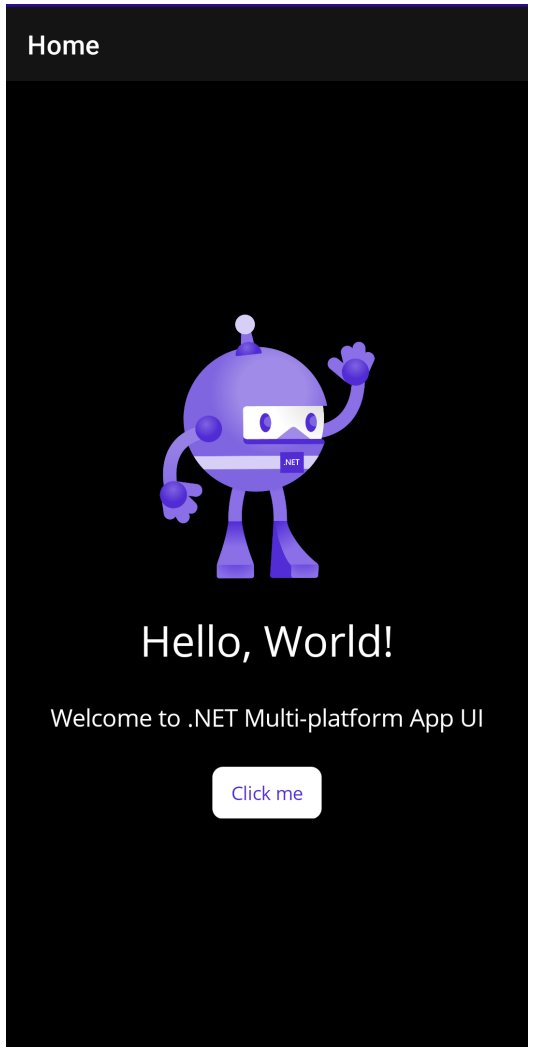
Indicates the focus order of the controls when a user is navigating via TalkBack/VoiceOver

```
<xct:SemanticOrderView x:Name="sov">
    <StackLayout>
                <Label x:Name="second" Text="Second control" />
                <Button x:Name="third" Text="Third control" />
                <Label x:Name="fourth" Text="Fourth control" />
                <Button x:Name="fifth" Text="Fifth control" />
                <Button x:Name="first" Text="First control" />
    </StackLayout>
</xct:SemanticOrderView>
```

```
sov.ViewOrder = new List<View> { first, second, third, fourth, fifth };
```

# App Theming

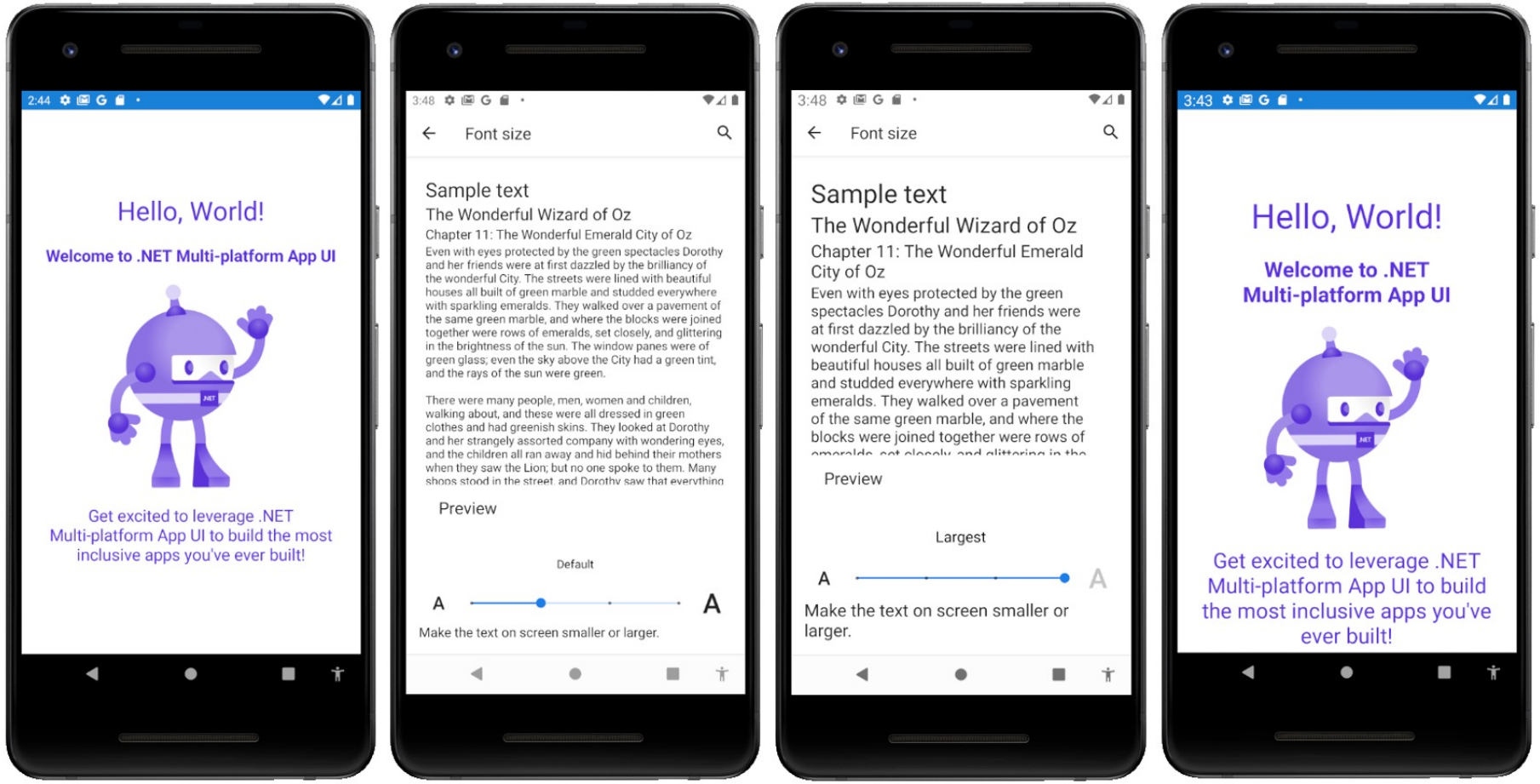



```
<Setter
    Property="BackgroundColor”
    Value="{AppThemeBinding
            Light={StaticResource White},
            Dark={StaticResource Black}}" />
```

# Font Autoscaling

## FontAutoScalingEnabled



# Built-in Accessibility

## Apply Accessibility traits to controls that have TapGestureRecogniz

Merged **PureWeen** merged 7 commits into `release/6.0.1xx-preview7` from `node_info_alternate` on 2 Aug 2021

Conversation 6 · Commits 7 · Checks 14 · Files changed 17



**PureWeen** commented on 30 Jul 2021 • edited — Member

**Description of Change**

If users attach a touch gesture recognizer that will activate with a single touch then we add accessibility traits that will inform readers that the element is clickable.

## Add traits to iOS Checkbox so it reads as Switch #1832

Merged **PureWeen** merged 4 commits into `main` from `ios_accessible_checkbox` on 27 Jul 2021

Conversation 0 · Commits 4 · Checks 14 · Files changed 2



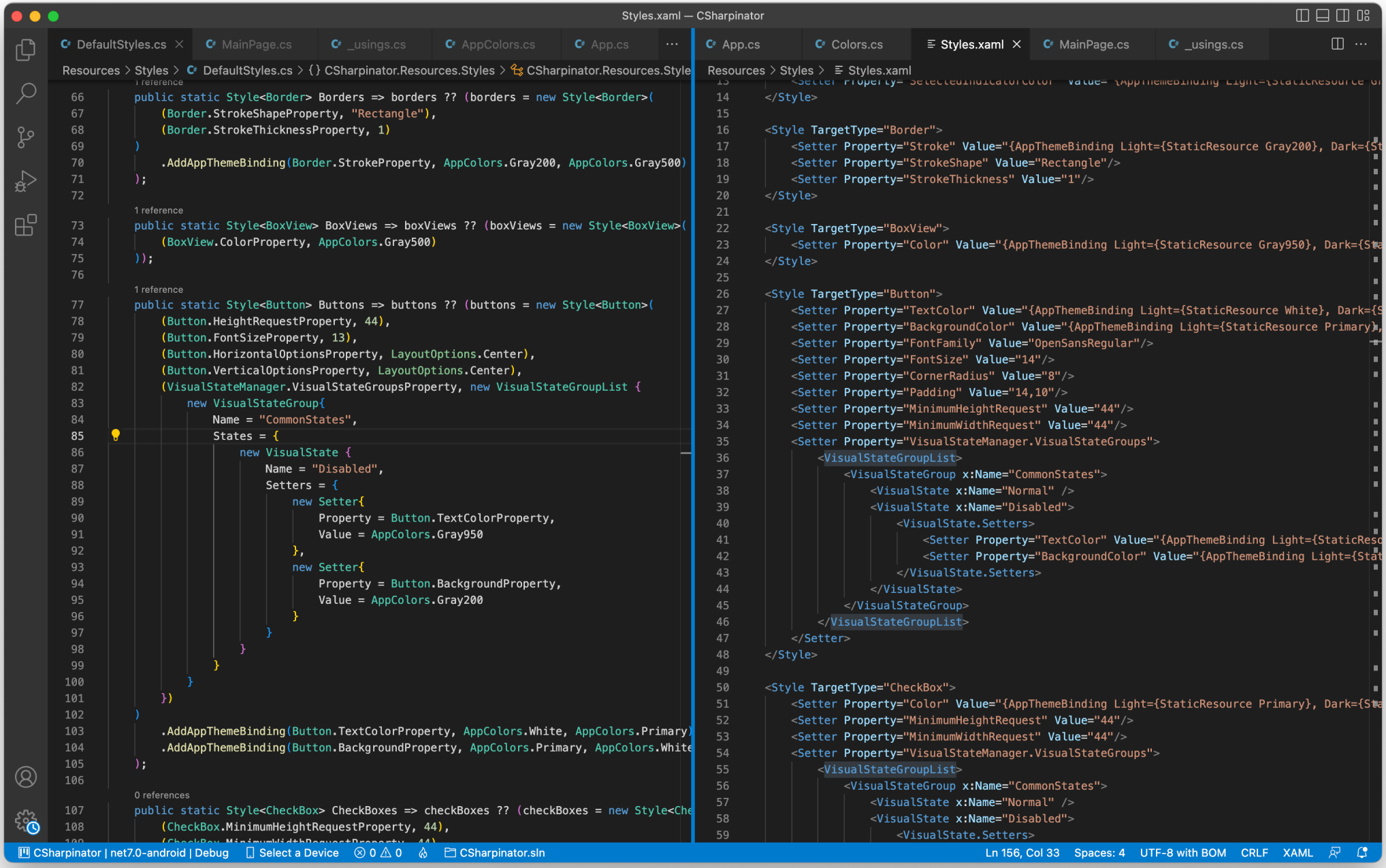
**PureWeen** commented on 27 Jul 2021 • edited — Member

**Description of Change**

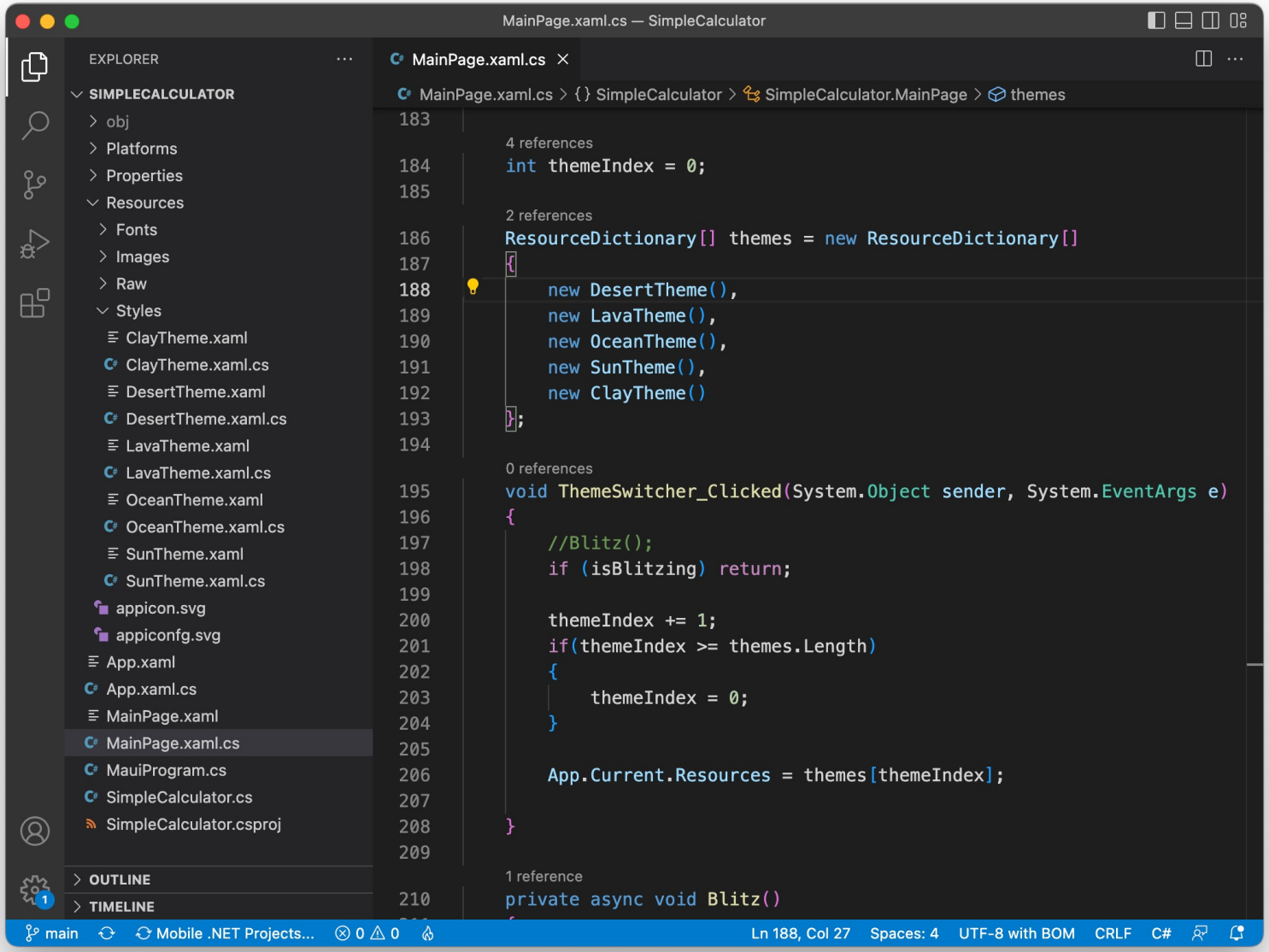
iOS doesn't natively have a checkbox so to make the current checkbox more accessible we will treat it semantically like a switch

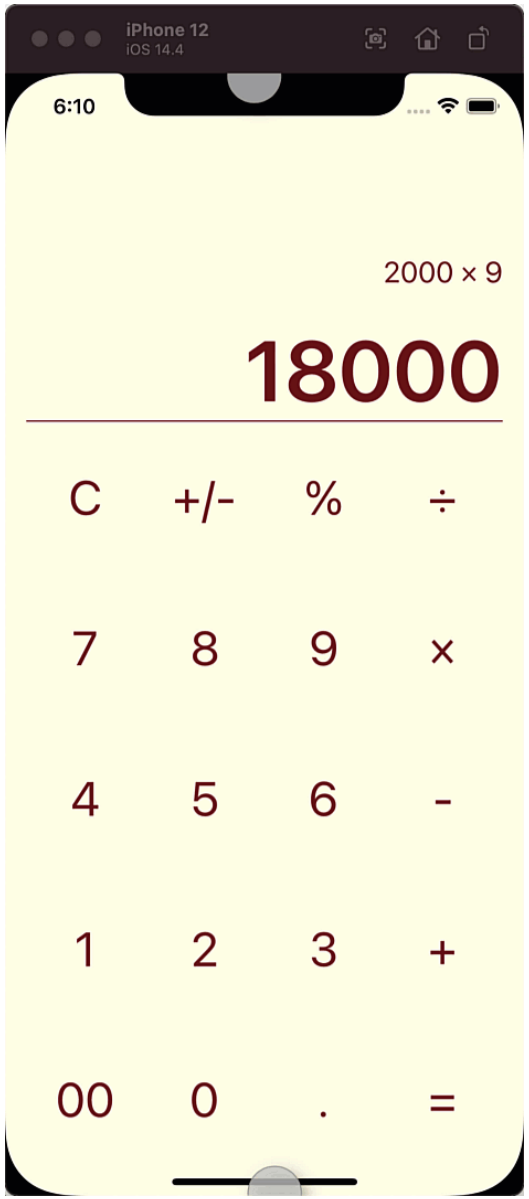
# Styles and Themes

# Styles


Styles.xaml — CSharpinator
DefaultStyles.cs
MainPage.cs
_usings.cs
AppColors.cs
App.cs
Colors.cs
Styles.xaml
Resources > Styles > DefaultStyles.cs > {} CSharpinator.Resources.Styles > CSharpinator.Resources.Style
Resources > Styles > Styles.xaml
CSharpinator | net7.0-android | Debug
Select a Device
CSharpinator.sln
Ln 156, Col 33
Spaces: 4
UTF-8 with BOM
CRLF
XAML

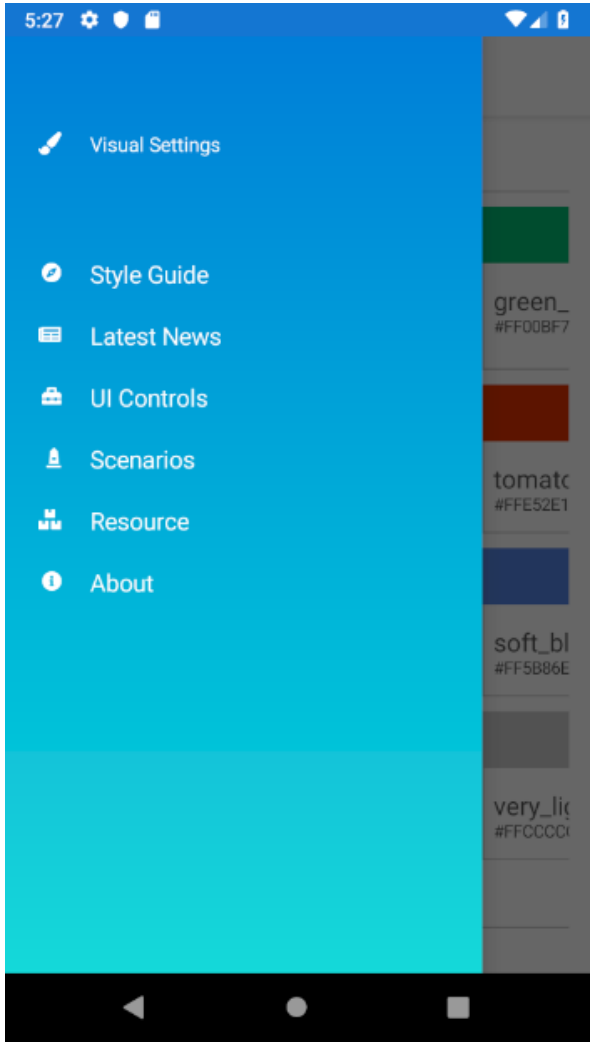
# Themes

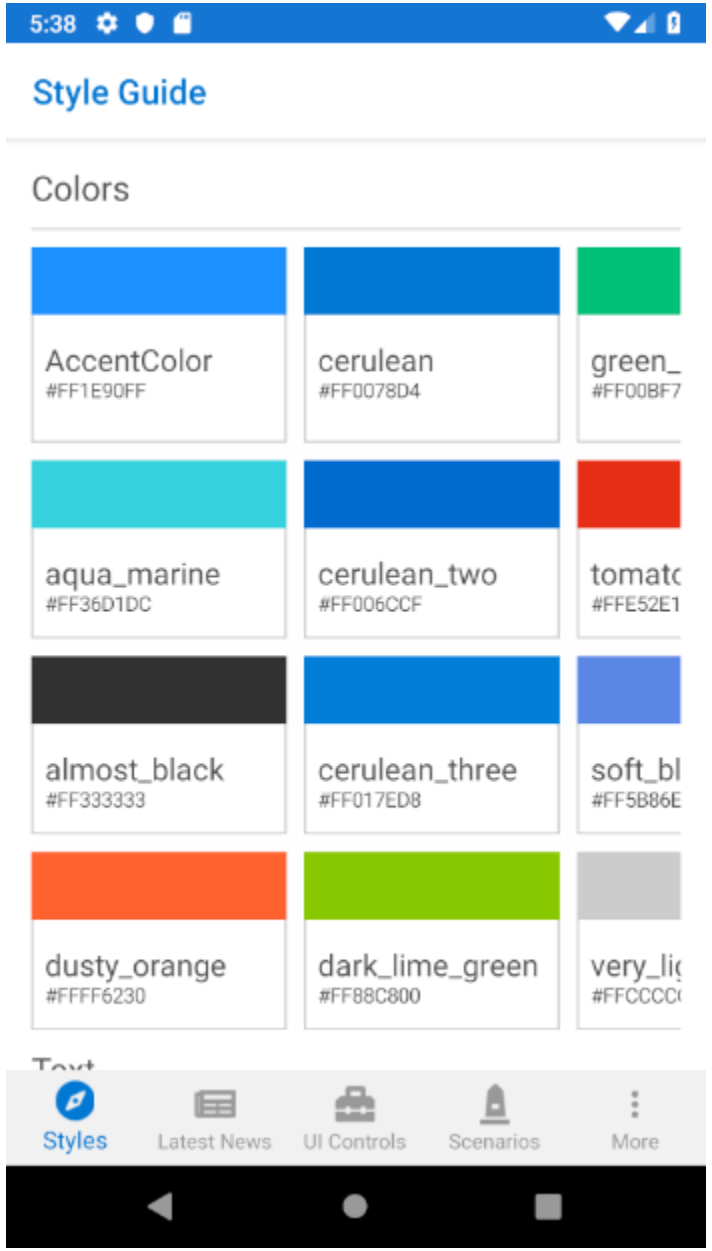

MainPage.xaml.cs — SimpleCalculator
EXPLORER
SIMPLECALCULATOR
obj
Platforms
Properties
Resources
Fonts
Images
Raw
Styles
ClayTheme.xaml
ClayTheme.xaml.cs
DesertTheme.xaml
DesertTheme.xaml.cs
LavaTheme.xaml
LavaTheme.xaml.cs
OceanTheme.xaml
OceanTheme.xaml.cs
SunTheme.xaml
SunTheme.xaml.cs
appicon.svg
appiconfg.svg
App.xaml
App.xaml.cs
MainPage.xaml
MainPage.xaml.cs
MauiProgram.cs
SimpleCalculator.cs
SimpleCalculator.csproj
OUTLINE
TIMELINE
MainPage.xaml.cs
MainPage.xaml.cs > { } SimpleCalculator > SimpleCalculator.MainPage > themes
4 references
int themeIndex = 0;
2 references
ResourceDictionary[] themes = new ResourceDictionary[]
{
new DesertTheme(),
new LavaTheme(),
new OceanTheme(),
new SunTheme(),
new ClayTheme()
};
0 references
void ThemeSwitcher_Clicked(System.Object sender, System.EventArgs e)
{
//Blitz();
if (isBlitzing) return;
themeIndex += 1;
if(themeIndex >= themes.Length)
{
themeIndex = 0;
}
App.Current.Resources = themes[themeIndex];
}
1 reference
private async void Blitz()
main
Mobile .NET Projects...
Ln 188, Col 27
Spaces: 4
UTF-8 with BOM
CRLF
C#

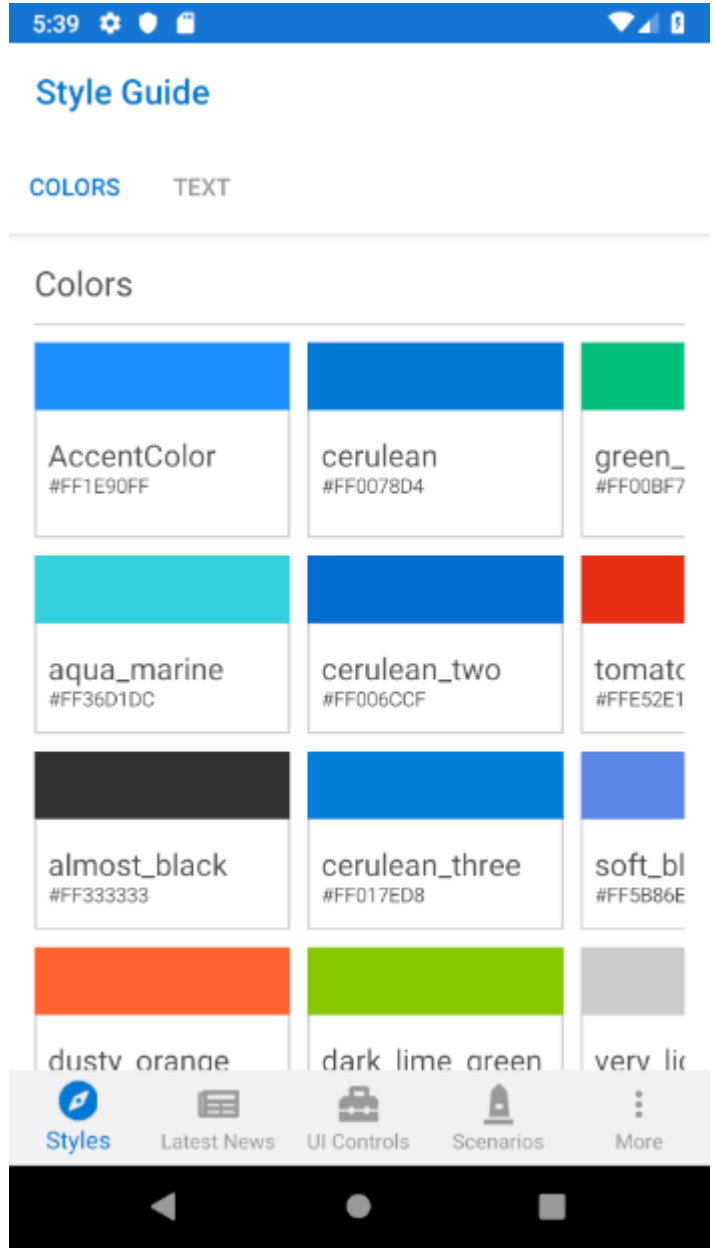

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# Navigation

Flyout, Tabs, and more


5:27
Visual Settings
Style Guide
Latest News
UI Controls
Scenarios
Resource
About


5:38
Style Guide
Colors
AccentColor
#FF1E90FF
cerulean
#FF0078D4
aqua_marine
#FF36D1DC
cerulean_two
#FF006CCF
almost_black
#FF333333
cerulean_three
#FF017ED8
dusty_orange
#FFFF6230
dark_lime_green
#FF88C800
Styles
Latest News
UI Controls
Scenarios
More


5:39
Style Guide
COLORS
TEXT
Colors
AccentColor
#FF1E90FF
cerulean
#FF0078D4
aqua_marine
#FF36D1DC
cerulean_two
#FF006CCF
almost_black
#FF333333
cerulean_three
#FF017ED8
Styles
Latest News
UI Controls
Scenarios
More

# Shell

# Tabs Made Simple

```
<TabBar>
    <Tab Title="Browse" Icon="tab_feed.png">
        <ShellContent ContentTemplate="{DataTemplate local:ItemsPage}" />
    </Tab>
    <Tab Title="About" Icon="tab_about.png">
        <ShellContent ContentTemplate="{DataTemplate local:AboutPage}" />
    </Tab>
</TabBar>
```

```
<TabbedPage x:Class="App5.Views.MainPage">

    <TabbedPage.Children>
        <NavigationPage Title="Browse">
            <NavigationPage.Icon>
                <OnPlatform x:TypeArguments="FileImageSource">
                    <On Platform="iOS" Value="tab_feed.png"/>
                </OnPlatform>
            </NavigationPage.Icon>
            <x:Arguments>
                <views:ItemsPage />
            </x:Arguments>
        </NavigationPage>

        <NavigationPage Title="About">
            <NavigationPage.Icon>
                <OnPlatform x:TypeArguments="FileImageSource">
                    <On Platform="iOS" Value="tab_about.png"/>
                </OnPlatform>
            </NavigationPage.Icon>
            <x:Arguments>
                <views:AboutPage />
            </x:Arguments>
        </NavigationPage>
    </TabbedPage.Children>

</TabbedPage>
```

# Flyout Made Simple

```
<FlyoutItem Title="Browse"
    <ShellCon
</FlyoutItem>
<FlyoutItem T
    <ShellCon
</FlyoutItem>
```

```
<MasterDetailPage xmlns="http://xamarin.com/schemas/2014/
    xmlns:x="http://schemas.microsoft.com/winfx/2
    xmlns:d="http:
    xmlns:mc="http
    mc:Ignorable="
    xmlns:views="
    x:Class="App6.
  <MasterDetailPage.Maste
    <views:MenuPage />
  </MasterDetailPage.Mast
  <MasterDetailPage.Deta
    <NavigationPage>
      <NavigationPage
        <OnPlatform
          <On Pla
        </OnPlatfor
      </NavigationPag
      <x:Arguments>
        <views:Iter
      </x:Arguments>
    </NavigationPage>
  </MasterDetailPage.Deta

</MasterDetailPage>
```

```
menuItems = new List<Hom
            {
                new Home
                new Home
            };

            ListViewMenu

            ListViewMenu
            ListViewMenu
            {
                if (e.Se
                    retu

                var id =
                await Ro
            };
```

```
public partial class MainPage : MasterDetailPage
    {
        Dictionary<int, NavigationPage> MenuPages = new Dictionary<int, NavigationPage>();
        public MainPage()
        {
            InitializeComponent();

            MasterBehavior = MasterBehavior.Popover;

            MenuPages.Add((int)MenuItemType.Browse, (NavigationPage)Detail);
        }

        public async Task NavigateFromMenu(int id)
        {
            if (!MenuPages.ContainsKey(id))
            {
                switch (id)
                {
                    case (int)MenuItemType.Browse:
                        MenuPages.Add(id, new NavigationPage(new ItemsPage()));
                        break;
                    case (int)MenuItemType.About:
                        MenuPages.Add(id, new NavigationPage(new AboutPage()));
                        break;
                }
            }

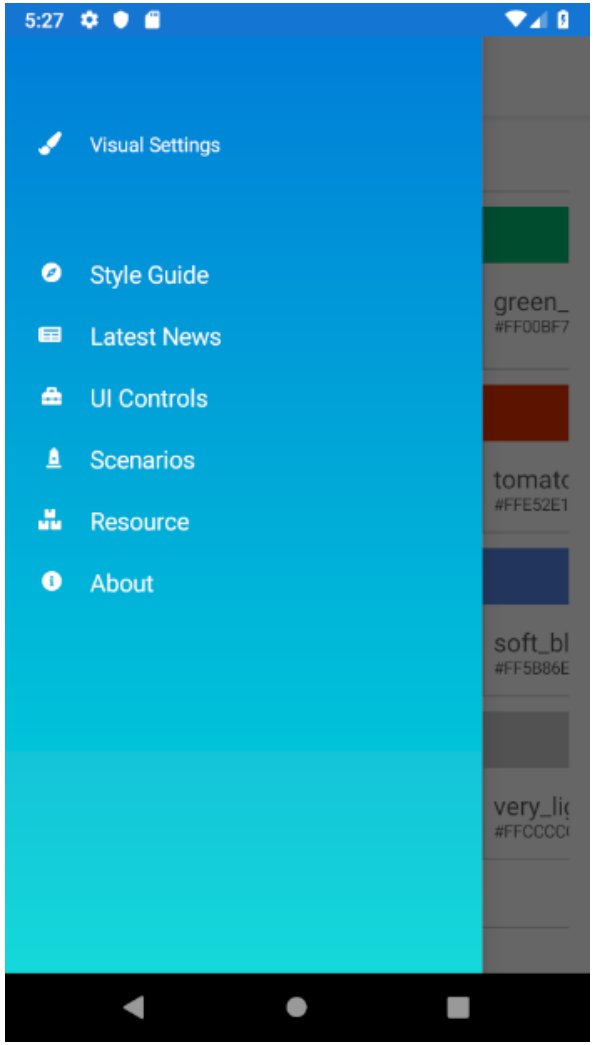
            var newPage = MenuPages[id];

            if (newPage != null && Detail != newPage)
            {
                Detail = newPage;

                if (Device.RuntimePlatform == Device.Android)
                    await Task.Delay(100);

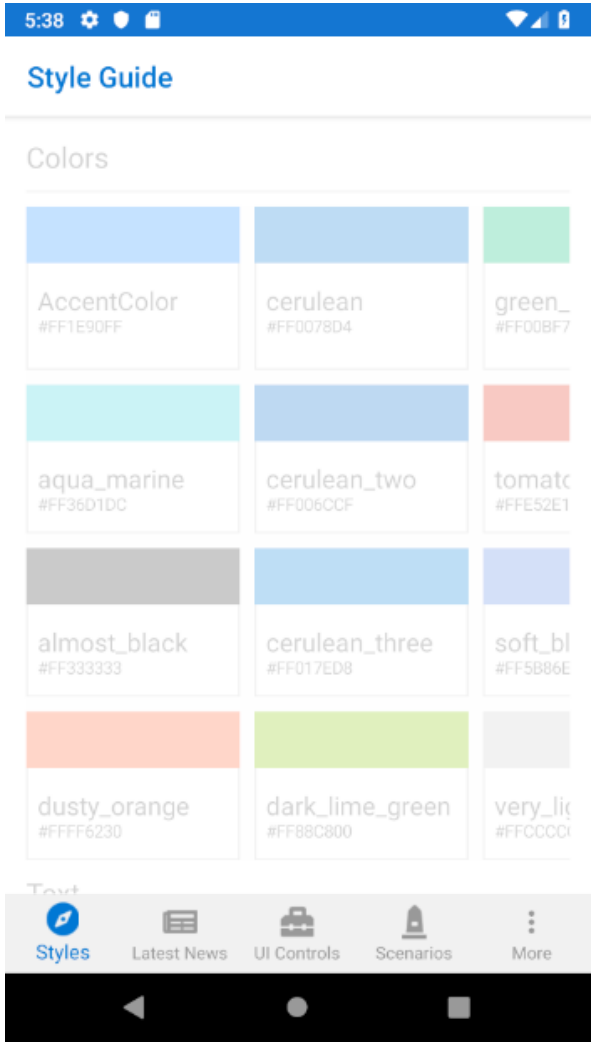
                IsPresented = false;
            }
        }
    }
```

```
</ContentPage>
```



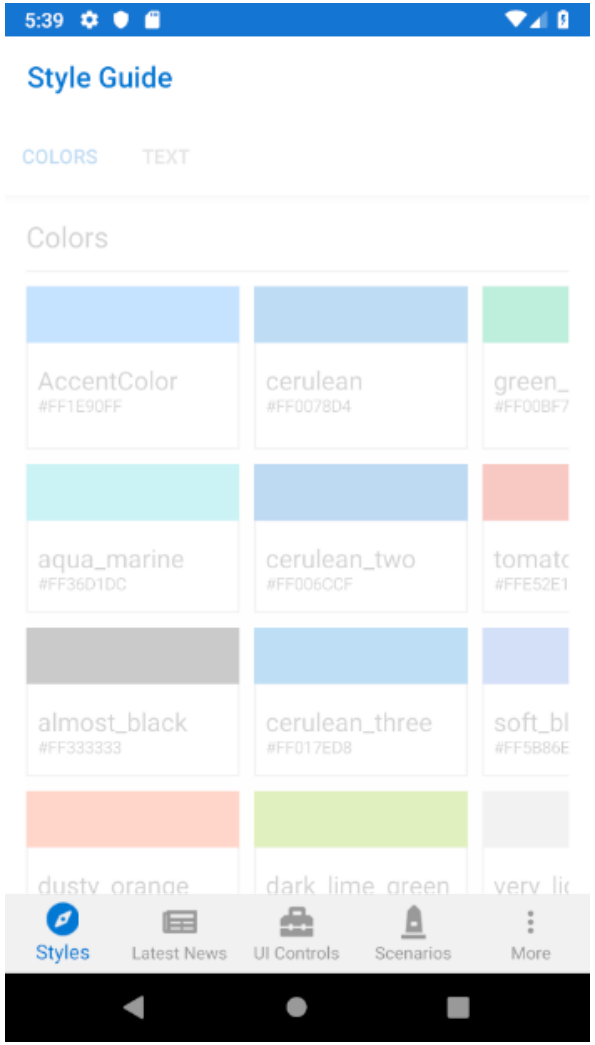

```
<Shell>
    <FlyoutItem
        Title="Style Guide"
        Icon="Compass.png">
        <ShellContent
            ContentTemplate="{DataTemplate p:StyleGuidePage}"/>
    </FlyoutItem>
</Shell>
```

# Shell




```
<Shell>
    <TabBar>
        <Tab
            Title="Styles"
            Icon="Compass.png">
            <ShellContent
                ContentTemplate="{DataTemplate
                                  p:StyleGuidePage}"/>
        </Tab>
    </TabBar>
</Shell>
```

# Shell



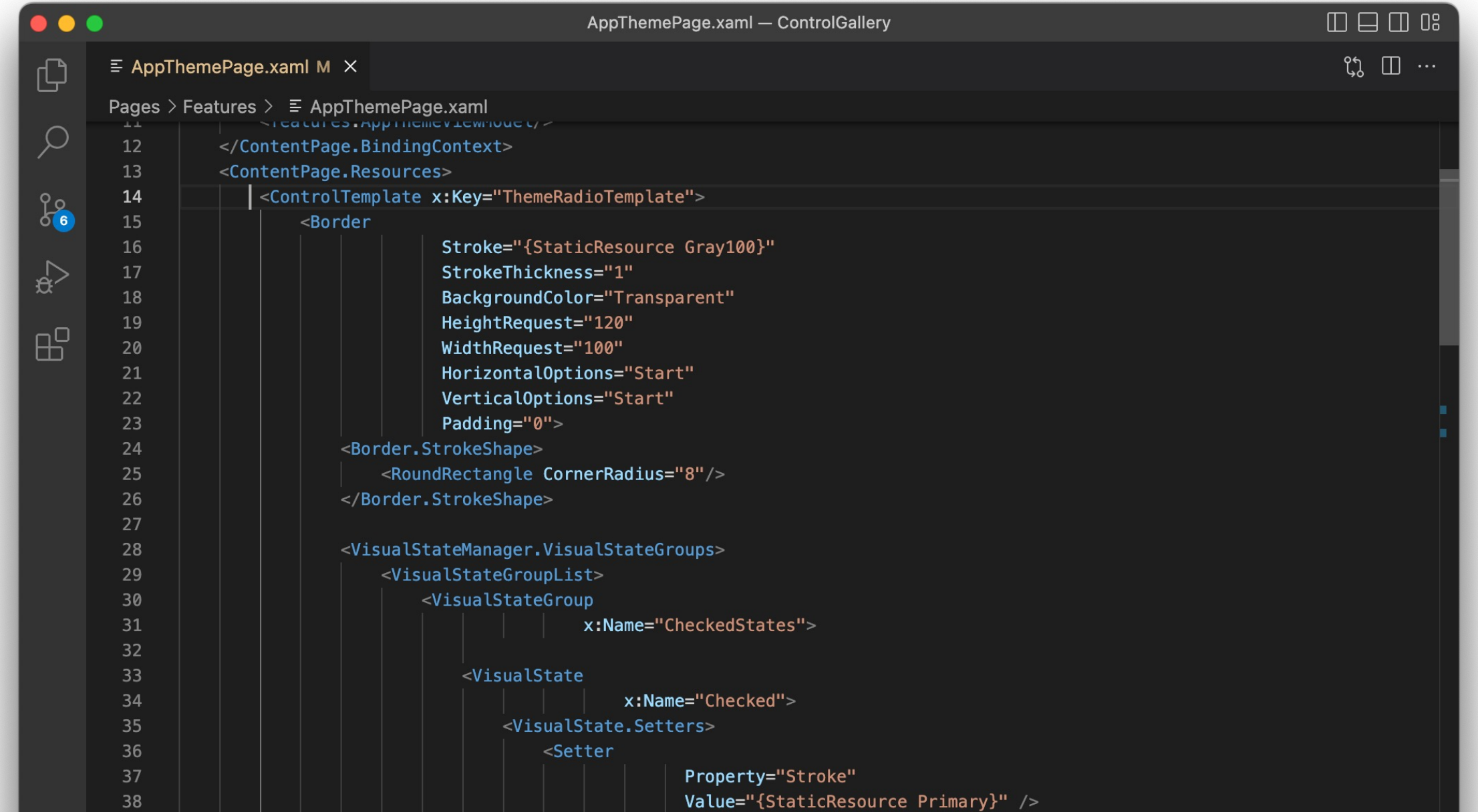

```
<Shell>
    <TabBar>
        <Tab
            Title="Style Guide"
            Icon="Compass.png">
            <ShellContent
                Title="Colors"
                ContentTemplate="{DataTemplate p:ColorsPage}"/>
            <ShellContent
                Title="Text"
                ContentTemplate="{DataTemplate p:TextPage}"/>
        </Tab>
    </TabBar>
</Shell>
```

# Shell

# Control Customizations

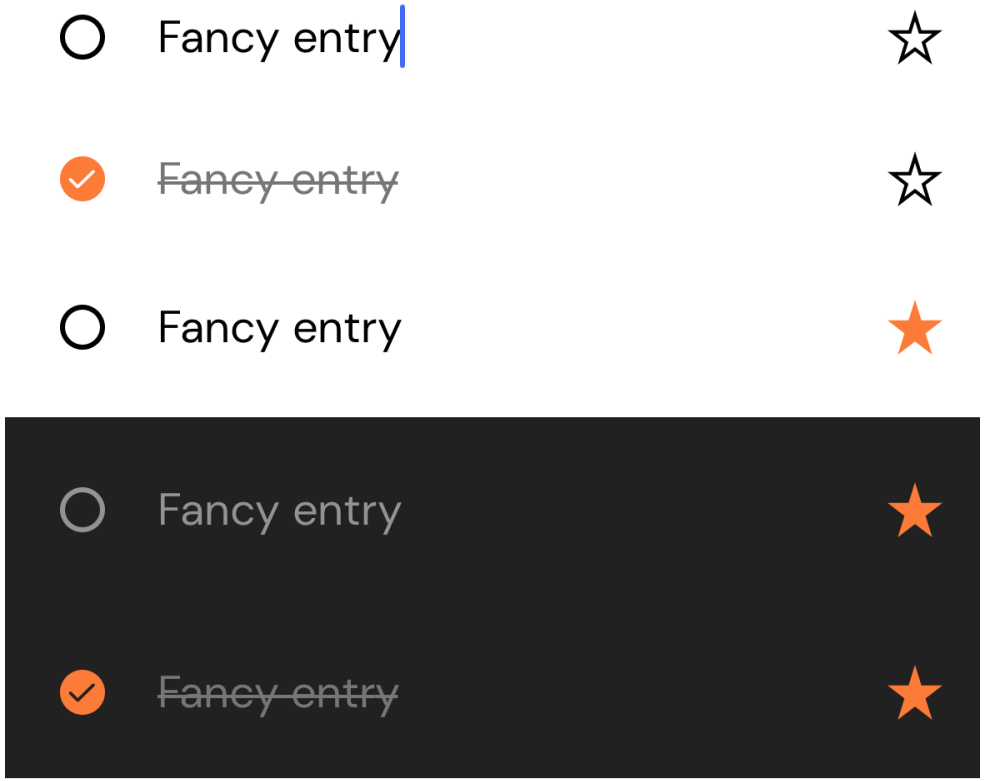
You have all the control

# Control Templating



# Control Composition

Grid
- Checkbox
- Entry
- Label
- ImageButton



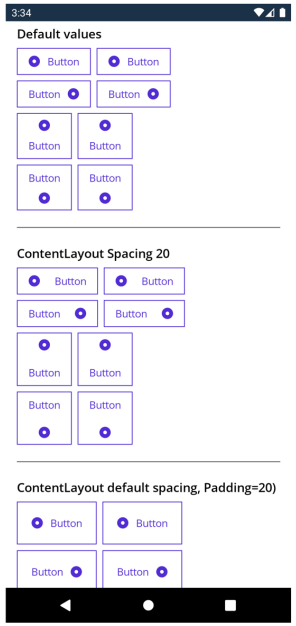

https://github.com/davidortinau/ToDoEntry/

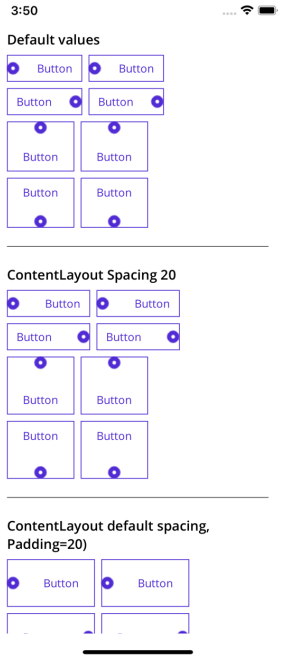
# Handlers

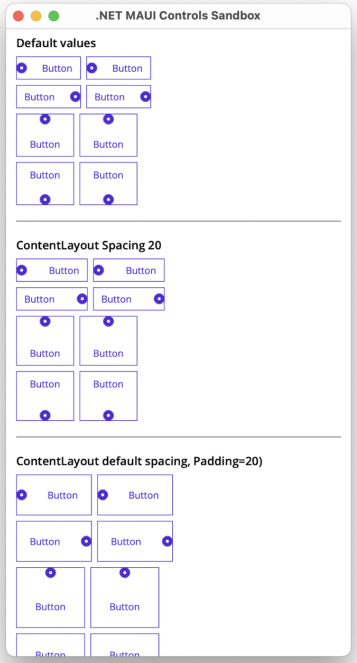
- **PrependToMapping**, which modifies the mapper for a handler before the .NET MAUI control mappings have been applied.
- **ModifyMapping**, which modifies an existing mapping.
- **AppendToMapping**, which modifies the mapper for a handler after the .NET MAUI control mappings have been applied.

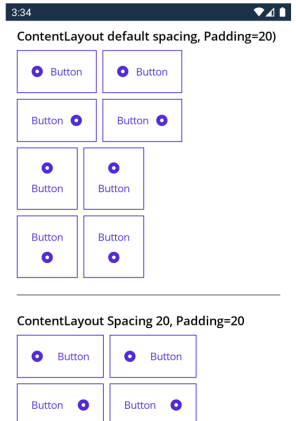
https://learn.microsoft.com/en-us/dotnet/maui/user-interface/handlers/customize

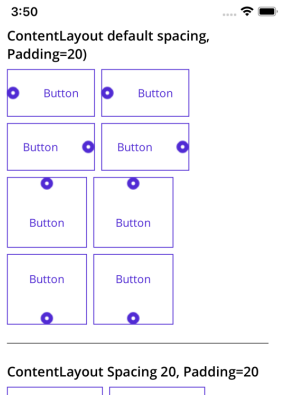
# A Bug's Story


3:34
Default values
Button
ContentLayout Spacing 20
ContentLayout default spacing, Padding=20)

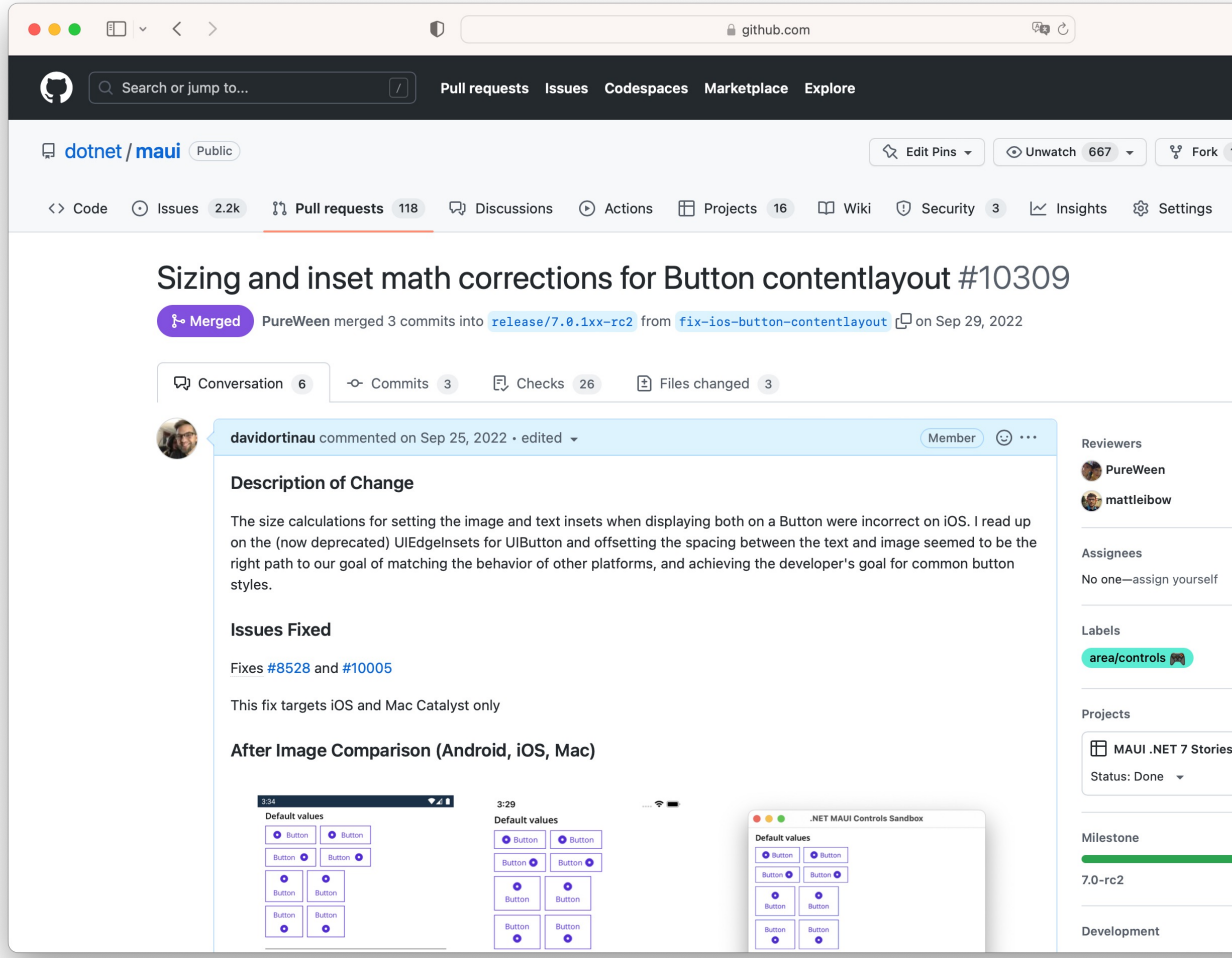

3:50
Default values
Button
ContentLayout Spacing 20
ContentLayout default spacing, Padding=20)


.NET MAUI Controls Sandbox
Default values
Button
ContentLayout Spacing 20
ContentLayout default spacing, Padding=20)


3:34
ContentLayout default spacing, Padding=20)
Button
ContentLayout Spacing 20, Padding=20

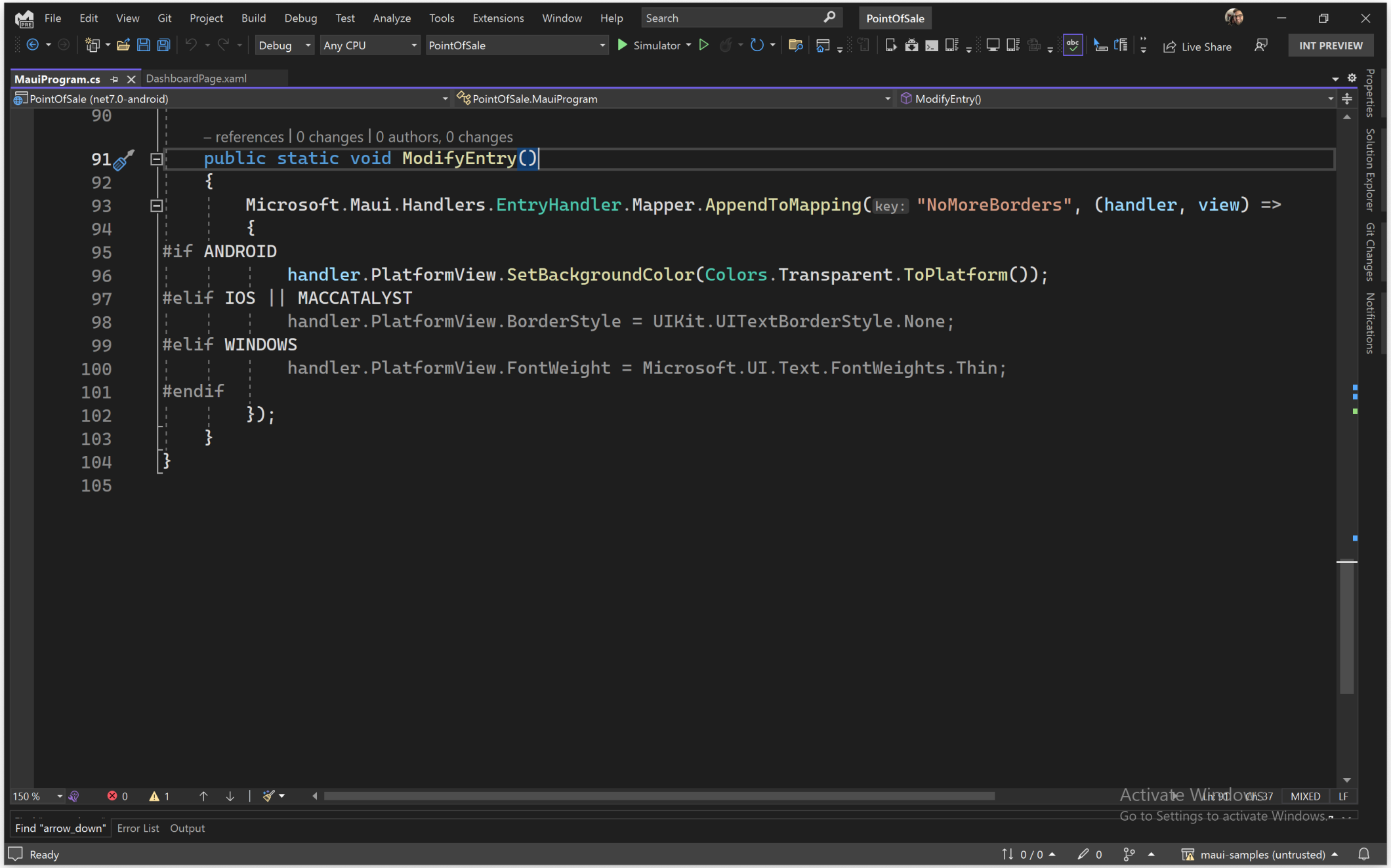

3:50
ContentLayout default spacing, Padding=20)
Button
ContentLayout Spacing 20, Padding=20


.NET MAUI Controls Sandbox
ContentLayout default spacing, Padding=20)
Button
ContentLayout Spacing 20, Padding=20


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Search or jump to...
Pull requests
Issues
Codespaces
Marketplace
Explore
dotnet / maui
Public
Edit Pins
Unwatch 667
Fork
Code
Issues 2.2k
Pull requests 118
Discussions
Actions
Projects 16
Wiki
Security 3
Insights
Settings
Sizing and inset math corrections for Button contentlayout #10309
Merged
PureWeen merged 3 commits into release/7.0.1xx-rc2 from fix-ios-button-contentlayout on Sep 29, 2022
Conversation 6
Commits 3
Checks 26
Files changed 3
davidortinau commented on Sep 25, 2022 • edited
Member
Description of Change
The size calculations for setting the image and text insets when displaying both on a Button were incorrect on iOS. I read up on the (now deprecated) UIEdgeInsets for UIButton and offsetting the spacing between the text and image seemed to be the right path to our goal of matching the behavior of other platforms, and achieving the developer's goal for common button styles.
Issues Fixed
Fixes #8528 and #10005
This fix targets iOS and Mac Catalyst only
After Image Comparison (Android, iOS, Mac)
Reviewers
PureWeen
mattleibow
Assignees
No one—assign yourself
Labels
area/controls
Projects
MAUI .NET 7 Stories
Status: Done
Milestone
7.0-rc2
Development

# A Bug's Story

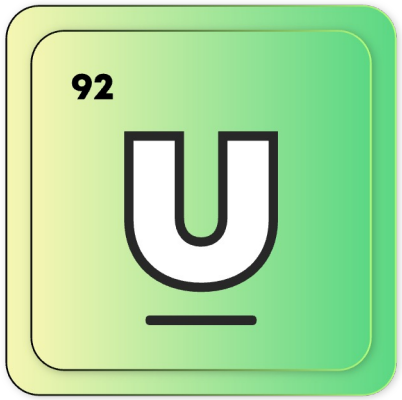
```
    1 reference
    private static void OverrideHandlers()
    {
#if IOS || MACCATALYST
        ButtonHandler.Mapper.ModifyMapping(nameof(Button.ContentLayout), (handler, btn, element) =>
        {
            UpdateContentLayout(handler.PlatformView, (Button)btn);
        });
#endif
    }
```

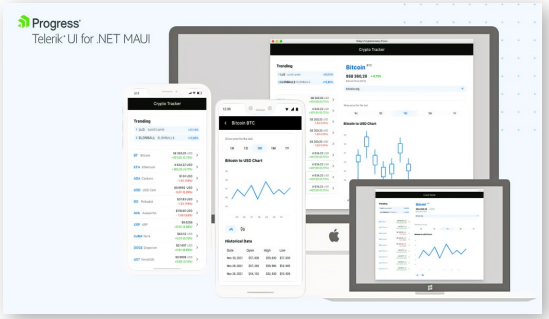

File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help
Search
PointOfSale
Debug
Any CPU
PointOfSale
Simulator
Live Share
INT PREVIEW
MauiProgram.cs
DashboardPage.xaml
PointOfSale (net7.0-android)
PointOfSale.MauiProgram
ModifyEntry()
– references | 0 changes | 0 authors, 0 changes
public static void ModifyEntry()
{
Microsoft.Maui.Handlers.EntryHandler.Mapper.AppendToMapping(key: "NoMoreBorders", (handler, view) =>
{
#if ANDROID
handler.PlatformView.SetBackgroundColor(Colors.Transparent.ToPlatform());
#elif IOS || MACCATALYST
handler.PlatformView.BorderStyle = UIKit.UITextBorderStyle.None;
#elif WINDOWS
handler.PlatformView.FontWeight = Microsoft.UI.Text.FontWeights.Thin;
#endif
});
}
}
150 %
0
1
MIXED
LF
Activate Windows
Go to Settings to activate Windows.
Find "arrow_down"
Error List
Output
Ready
0 / 0
0
maui-samples (untrusted)
Properties
Solution Explorer
Git Changes
Notifications

# Other Options

- **Draw** your own UI with Microsoft.Maui.Graphics
- **Choose** from any of the awesome control libraries in the .NET ecosystem.

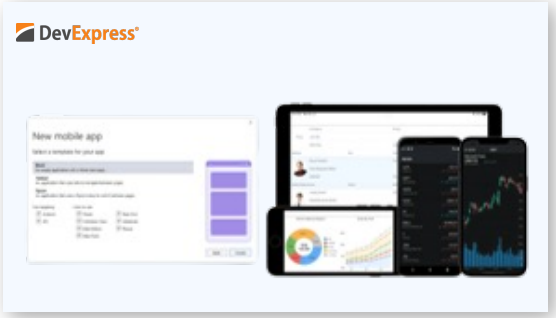






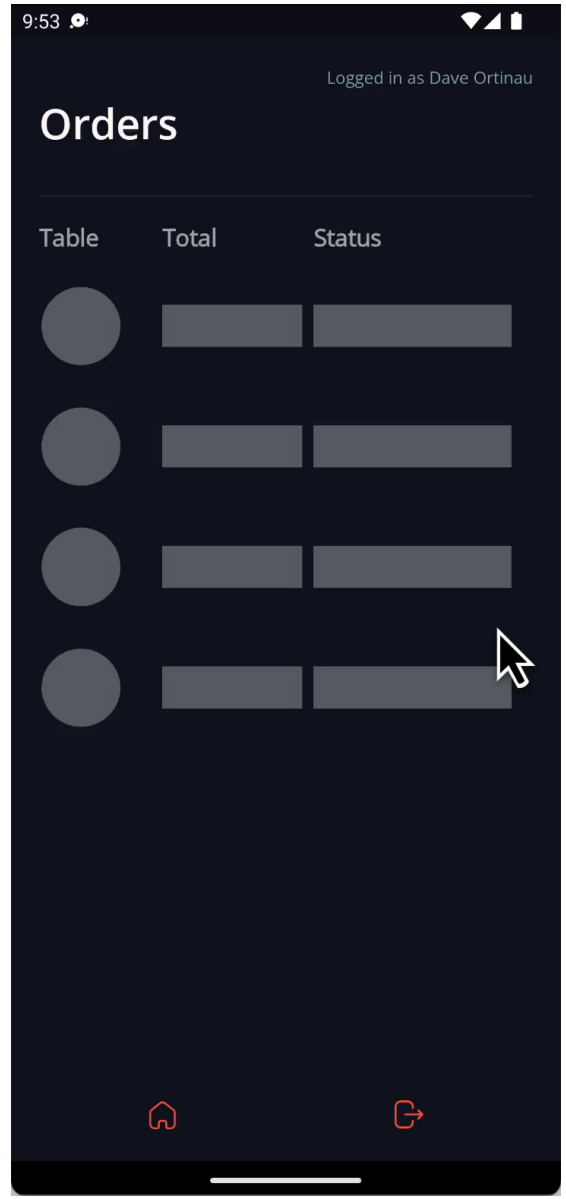





# State and Animation

# StateContainer

- Show placeholders while you are loading content
- More meaningful than indeterminant spinners
- BoxView is your friend
- Part of .NET MAUI Community Toolkit
- Animations courtesy of AlohaKit.Animations




Show me a demo!

# Additional Resources

- https://snppts.dev
- https://github.com/jsuarezruiz/awesome-dotnet-maui
- https://github.com/dotnet/maui-samples
- https://github.com/davidortinau/

# Thanks! Let’s talk.

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